CoviCare: Tracking Covid-19 using PowerBI

Kunal Chakate
Department of Infromation Technology
Symbiosis Institute of Technology
Pune,India
kunal.chakate.btech2019@sitpune.edu.in

Govinda Giri
Department of Infromation Technology
Symbiosis Institute of Technology
Pune,India
govinda.giri.btech2019@sitpune.edu.in

Sudhanshu S. Gonge
Department of Infromation Technology
Symbiosis Institute of Technology
Pune,India
sudhanshu.gonge@sitpune.edu.in

Aditya Deshpande
Department of Infromation Technology
Symbiosis Institute of Technology
Pune,India
aditya.deshpande.btech2019@sitpune.edu.in

Yogeshwar Pawade
Department of Infromation Technology
Symbiosis Institute of Technology
Pune,India
yogeshwar.pawade.btech2019@sitpune.edu.in

Rahul Joshi
Department of Infromation Technology
Symbiosis Institute of Technology
Pune,India
rahulj@sitpune.edu.in

2022 8th International Conference on Signal Processing and Communication (ICSC) | 978-1-6654-5430-8/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICSC56524.2022.10009069

***Abstract*— In late 2019, the world witnessed the emergence of one of the most alarming pandemics recorded in the Contemporary Age, Covid-19. The new coronavirus is responsible for causing the disease Covid-19 and it has since spread to over 180 countries. As the disease spreads over the world, it has become a global pandemic, threatening global public health, and presenting a huge threat to global civilization. To oppose and prevent the spread of COVID-19, everyone should be well informed on the disease's constantly changing status. To accomplish this purpose, a COVID-19 analytical tracker was developed using PowerBI to provide the most up-to-date sickness status as well as critical analytical insights. The Covid tracker is intended for the general public who lack specialized statistical knowledge. It tries to express insights using a variety of basic and succinct data visualizations backed up by reputable data sources. The purpose of this paper is to describe the key strategies used to create the insights presented on the tracker, which include data retrieval, normalization techniques, and ML/DL models. In addition to explaining the facts and justifications for the approaches used, the paper includes several major discoveries made in relation to COVID-19 utilizing the methodologies.**

Keywords*— *COVID-19, CovidTracker, Common Symptoms, Data Visualization, Hypothesis Testing, Normalization, ML/DL models.

I. Introduction

COVID-19 tracker features include odometers for the most recent status of COVID-19 cases, trend analysis, and projection of COVID-19 cases, useful visualizations of the most prevalent symptoms and risk factors, and patient demographic distributions. We have included many more features such as a Hospital Management System and also a real-time encrypted messenger for doctors and patients to chat.

This application is also a real time tracker where the user can see the live updates of the covid cases. For prediction machine learning/deep learning models are used. ANN and LSTM models are used for prediction and analysis. Bulit forecasting model for find coronaviruses cases count and Our Model is going to run with LSTM model.

With only a few lines of code, we can easily do an exploratory data analysis using the open source Python package called Pandas profiling.

Following that, there will be a brief overview of each key feature, a discussion of the important methodology behind the feature, and a summary of selected findings from the feature.

II. Literature Survey

Nikita Sharma et.al.(2022), in [1] Explains project's design and architecture for healthcare data analytics using PowerBi.

Kotapati Saimanoj et.al.(2020), in [2] is used to define the types of jobs that need to be completed and managed in Hospital management system utilising Web technology.

Daniel Matthias et.al.(2021), in [3] was formulated for Data analytics and visualization using PowerBI.

Arnela Salkić et.al.(2020), in [4] defined Covid-19 data analysis-predicting patient recovery by characterising the landscape of covid-19 themed cyberattacks and defences and few tiny details were deduced.

l..Wang et.al.(2015), in [5]Conventional data visualization methods as well as the extension of some conventional methods to Big Data applications are introduced in this paper. The challenges of Data visualization are discussed. New methods, applications, and technology progress of Data visualization is presented.

Bartosz Sawik et.al.(2022), in [6] analysed currently available solutions that helps to monitor the global epidemiological situation, including travel restrictions, as well as proposing a new solution dedicated to users who want to keep updated with the current restrictions and Covid-19 related statistics.

Mir Mehedi Ahsan Pritom et.al(2021), in [7] characterized landscape of Covid 19 themed cyberattacks and defenses.

Eysenbach,G. et.al (2009), in [8] Infodemiology and infoveillance: framework for an emerging set of public health informatics methods to analyse search, communication and publication behaviour on the internet.

Yusuf Perwej et.al (2017), in [9] analysed the experiential study of Big Data International transaction of systems.

Manohar V et.al(2018), in [10] Defined Tableu: a high throughput and predictable VM scheduler for high density workloads.

Pascual-cid Victor, et.al(2015) in [11] Explained Information visualization system for the understanding of web data.

L.wang et.al (2015), in [12] visualised big data challenges and technology progress. S.K.Khor et.al(2020), in [13] Stated transparency record of covid-19 in Malaysia.

J.Passy et.al.(2020), in [14] stated why U.S government's effort to contain covid outbreak was never going to be successful on market watch.

D.E.Bloom et.al.(2018), in [15] defined New and resurgent infectious diseases can have far reaching economic repercussions on finance and development.

O'Neill et.al.(2020), in [16] analysed a flood of coronavirus apps are tracking citizens and its time to track them.

Ahmed et.al.(2020), in [17] surveyed Covid-19 contact tracing apps.

Bareinboim et.al.(2016) in [18] proposed and solved casual inference and data fusion problem.

Banker, S.Park et.al.(2020) in [19] Evaluated prosocial covid-19 messaging frames: evidence from a field study on facebook.

Sohrabi et.al(2020) in [20] explained how World health organizations declared global emergency: a review of the 2019 novel coronavirus(Covid-19).

III. OBJECTIIVE

The Power-Bi live dashboard is a basic, simple application that will provide access to a large number of data related to Covid-19. It consists of data featuring the number of victims suffering from corona. The percentage of people that have been vaccinated. The number of people who lost their lives due to this pandemic, and a lot more.

Users can search data according to their needs with the help of different filters available along with the search button.

The dashboard as well as its dataset are dynamic tools that will be updated, altered, and rectified as needed.

IV. PROPOSED SYSTEM AND SOLUTION

To power our CoviCare Tracker, we have been monitoring multiple reputable data sources such as Johns Hopkins University (JHU) and Worldometer. Generally, we've seen that Worldometer updates more quickly than JHU and uses their data for the current day. After some cleaning, we use JHU as our historical book of record. We are happy to offer most data from the Tracker to our customers and others.

- Embed any of these live and interactive components using an iFrame.
- Explore and export key Coronavirus metrics with CoviCare's Covid-19 Data Explorer.
- Access CoviCare's curated Covid-19 cases and vaccinations Datasets.
- Instantly create visualizations from CoviCare's curated Covid-19datasets.
- Consult Doctors online with our online consultation system

V. METHODODLOGY AND WORKFLOW

This procedure connects to API to acquire updated data on the COVID-19 virus's propagation. Confirmed cases and fatalities, as well as vaccination time data, are represented in various ways per country. The dashboard may be seen locally using the open-source PowerBI Analytics Platform desktop interface, as well as published on Firebase and built with ReactJS.

Algorithms used:

ANN:
Artificial neural networks are processing information that draw inspiration from the biological neural networks that make up animal brains. These systems are frequently referred to as neural networks or neural nets. Artificial neurons, which loosely replicate the neurons in a biological brain, are a group of linked units or nodes that form the foundation of an ANN.

LSTM:
An enhanced RNN, or sequential network, called a long short-term memory network, permits information to endure. It is capable of resolving the RNN's vanishing gradient issue. RNNs, sometimes referred to as recurrent neural networks, are utilised for permanent memory.
Due to diminishing gradient, RNN have the flaw of being unable to recall long-term dependencies. Long-term dependence issues are specifically avoided while designing LSTMs.

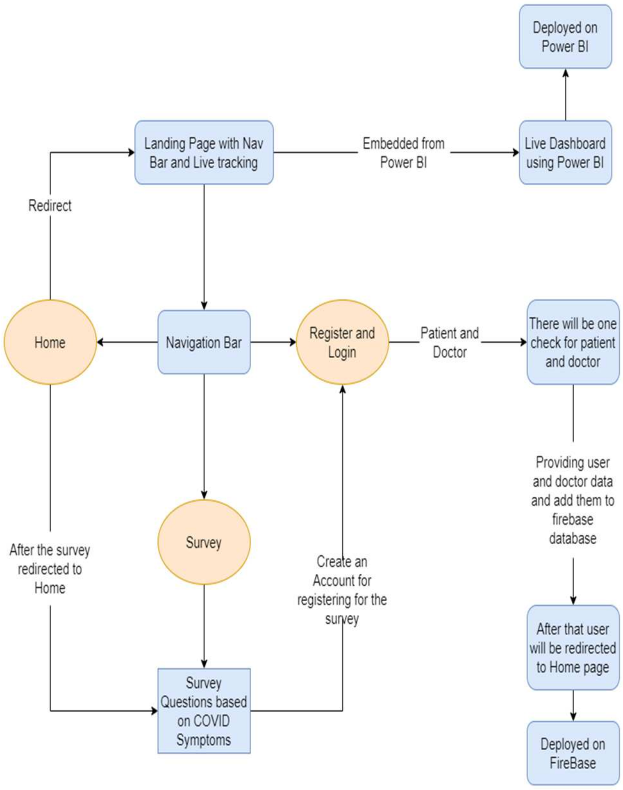

Figure 1. Workflow Diagram

1) Home

2) In the Home page has a nav bar which consists of Live Tracking button, Register and Login button, Survey button, home button.

3) After clicking on Live tracking button, one will be redirected to a page which embedded is with Power BI which is a Live Dashboard for covid cases.

4) After clicking on Register/Login button there will be to options Doctor Register/Login , Patient Register/Login.

6) The user has to provide Patient/Doctor right credentials to access the existing account they have to sign up (Register)

using proper credentials which will be added to a firebase database.

7) After that User will be redirected to the home page.

8) After clicking on the Survey button, the User will be redirected to a page with Survey Questions based on COVID Symptoms.

9) The User should create an account for registering for the Survey.

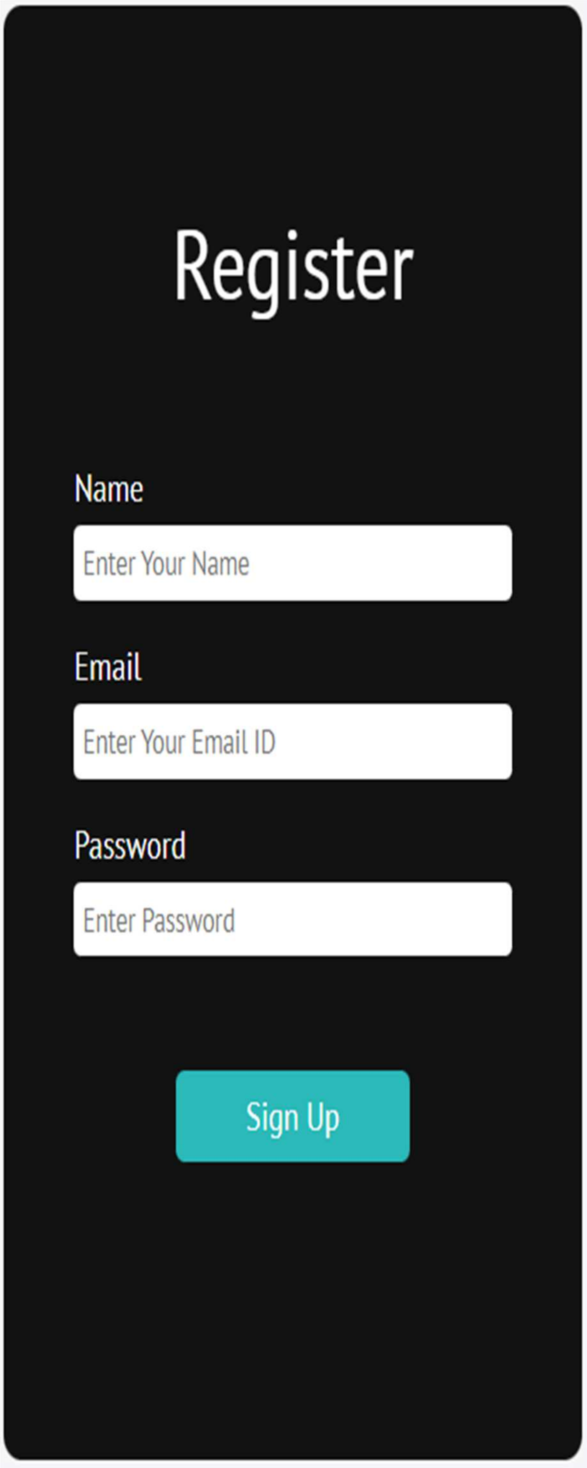

Figure 2. Registration Dashboard

If a user does not have an account, he or she can make that using that can create an account. A registration page for the User (i.e., Patient)

VI. RESULT AND DISCUSSION

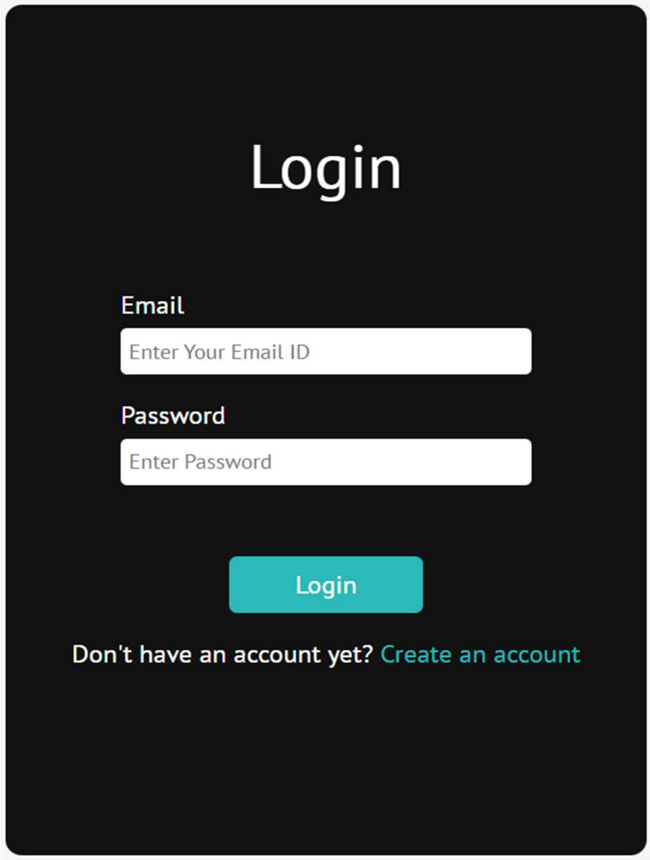

Figure 3. Doctor/Patient Login

Doctor/Patient Login page consists of features like a username textbox, a password textbox, and a sign-in button are included in the Sign-in window. If the user provides wrong credentials the system will give a prompt "Invalid username and password".

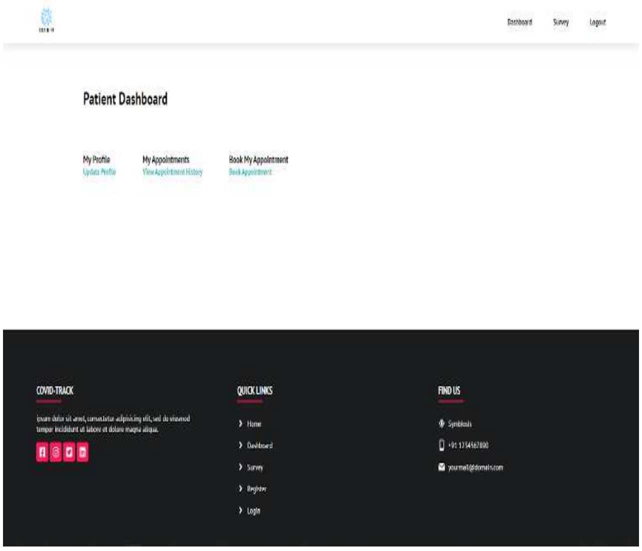

Figure 4. Patient Dashboard

The Patient Dashboard page consists of features like a My profile button, a My appointments button, and a book my appointment button are included in the main window. In addition to all of this, there is a navigation bar where there are three buttons: Dashboard button, Book appointment button, appointment history button. Also, there is a navigation bar where there is toggle button and user button.

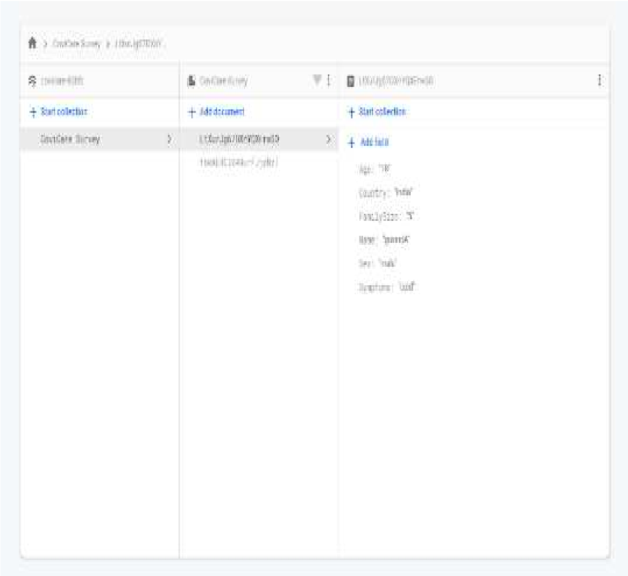

Figure 5. Firebase Database

If a user creates an account, all the information which he or she provides will be stored in this database. Firebase database is also being used for storing the credentials and survey details.

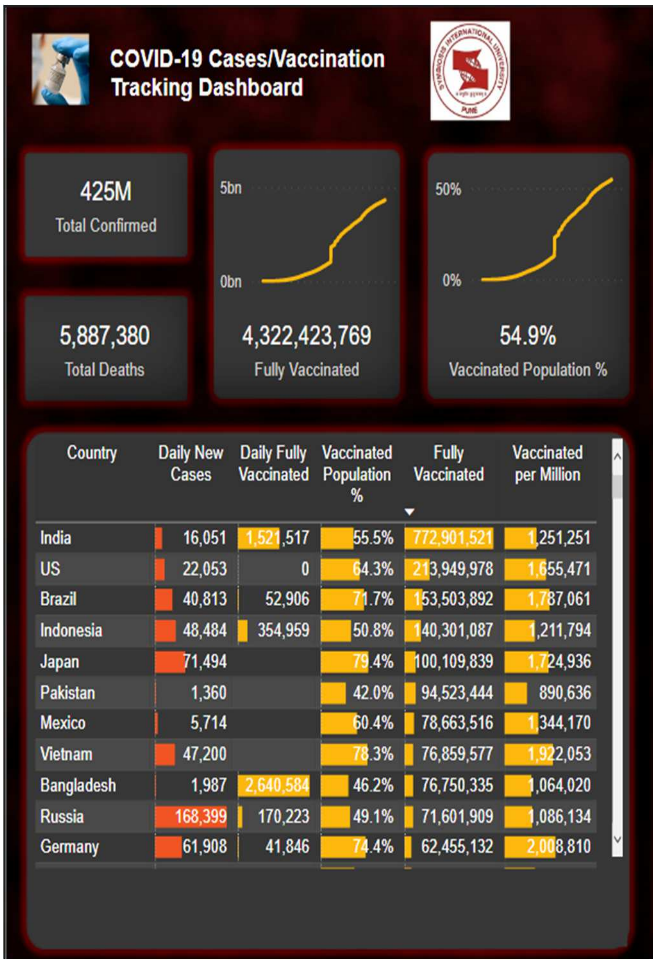

Figure 6.1 Covid-19 Dashboard

A dashboard that displays the current state of COVID-19 and related information. Daily and cumulative views Absolute departure as a percentage of the previous day. Date, continent, nation, and state are all options.

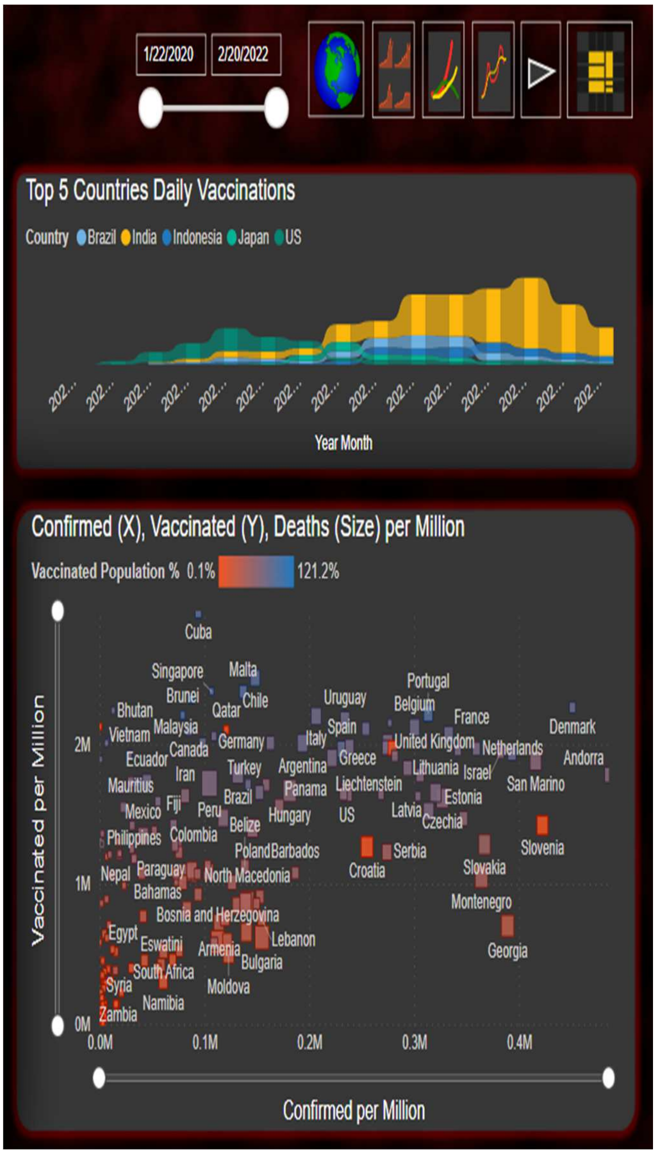

Figure 6.2 Covid-19 Dashboard

VII. Conclusions

The COVID-19 tracker was used to show current COVID-19 trends in various regions, as well as COVID-19 symptoms and patient demographics. The research also looks into the COVID-19 tracker's procedures, such as data retrieval, data transformation and standardization, and the ML/DL model. However, we must be cautious in accepting the conclusions because there may be data quality issues, such as a large percentage of missing data and erroneous patient-level data entries. We can try reproducing the gained insights to corroborate the findings of this study if we get access to an updated dataset near the end of the epidemic. During a global outbreak like COVID-19, it is vital for the public to have access to the most up-to-date information. During a worldwide pandemic like COVID-19, it's critical for the public to know the latest state of the epidemic and to be well-informed on essential disease insights. A tool like a COVID-19 tracker will aid the public in disseminating accurate and trustworthy information on COVID-19's spread. The tracker's research and effort are driven by a social obligation to educate the general public by offering scientifically based data analysis, predictions, and pertinent discoveries. This study and research effort are still in progress since there are many more COVID-19 experiments that can be conducted.

REFERENCES

[1] Nikita Sharma and Dhrubashish Sarkar. (2022) "Healthcare Data Analytics Using Power BI". IEEE [Online] Available from: DOI: 10.4018/IJSI.293267

[2] Kotapati Saimanoj, Grandhi Poojitha, Khushbu Devendra Dixit, Laxmi Jayannava. (2020) "Hospital Management System utilising WebTechnology". IEEE [Online]. "(PDF) Hospital Management System using Web Technology (researchgate.net)".

[3] D. Matthias, C. Managwu (2021). Data analytics and visualization using Power BI and MS Excel for COVID-19 (Coronavirus). IEEE [Online] Available from: DOI: 10.13140/RG.2.2.25204.48001

[4] Arnela Salkić, Nermina Durmić. (2020) "COVID-19 Data Analysis – Predicting Patient Recovery". IEEE [Online] Available from: DOI: 10.38124/IJISRT20AUG180

[5] Mir Mehedi Ahsan Pritom, Kristin M. Schweitzer, Raymond M. Bateman, Min Xu, Shouhuai Xu. (2021) "Characterizing the Landscape of COVID-19 Themed Cyberattacks and Defenses" arXiv:2102.13223[cs.CR]. Available from: https://doi.org/10.48550/arXiv.2102.13223

[6] Pandemic (COVID-19)". Published online at OurWorldInData.org. Retrieved from: Culotta, A. Towards detecting influenza epidemics by analyzing Twitter messages. In Proc. of the 1st Workshop on Social Media Analytics, 115–122, "https://doi.org/10.1145/1964858.1964874" (2010).

[7] Polgreen, P. M., Chen, Y., Pennock, D. M., Nelson, F. D. & Weinstein, R. A. Using internet searches for influenza surveillance. Clin. Infect. Dis. 47, 1443–1448 (2008).

[8] Eysenbach, G. Infodemiology and infoveillance: framework for an emerging set of public health informatics methods to analyze search, communication and publication behavior on the internet. J. Med. Internet Res. 11, "https://doi.org/10.2196/jmir.1157" (2009).

[9] Yusuf Perwej. (2017). "An Experiential Study of the Big Data," International Transaction of Electrical and Computer Engineers System (ITECES), USA, ISSN (Print): 2373-1273 ISSN (Online): 2373-1281, Science and Education Publishing, 4(1),14-25.

[10] Manohar V, Arpan G and Björn B. (2018). Brandenburg. 2018. Tableau: A High- Throughput and Predictable VM Scheduler for High-Density Workloads. In EuroSys '18: Thirteenth EuroSys Conference, ACM, New York, NY, USA,16.

[11] Pascual-cid Victor. (2008). "An Information Visualization System for the Understanding of Web Data", Proceedings of IEEE Symposium on Information Visualization (INFOVIS).

[12] L. Wang, G. Wang, and C.A. Alexander. (2015). "Big data and visualization: methods challenges and technology progress", Digital Technologies, 1(1), 33-38.

[13] S. K. Khor, "Malaysia does not have a good record of transparency," The Star, 15 January 2020. [Online].

[14] J. Passy, "Here's why the U. S. government's effort to contain the coronavirus outbreak was never going to be successful," Market Watch, 29 February 2020. [Online].

[15] D. E. Bloom, D. Cadarette and JP Sevilla, "New and resurgent infectious diseases can have far-reaching economic repercussions," Finance and Development, vol. 55, no. 2, pp. 46-49, 2018.

[16] O'Neill, P. H., Ryan-Mosley, T. & Johnson, B. A flood of coronavirus apps are tracking us. Now it's time to keep track of them. MIT Technology Review (7 May 2020).

[17] Ahmed, N. et al. A survey of COVID-19 contact tracing apps. IEEE Access 8, 134577–134601 (2020).

[18] Bareinboim, E. & Pearl, J. Causal inference and the data-fusion problem. Proc. Natl Acad. Sci. USA 113, 7345–7352 (2016).

[19] Banker, S. & Park, J. Evaluating prosocial COVID-19 messaging frames: evidence from a field study on Facebook. Judgm. Decis. Mak. 15, 1037–1043 (2020).

[20] Sohrabi, C. et al. World Health Organization declares global emergency: a review of the 2019 novel coronavirus (COVID-19). Int. J. Surg. 76, 71–76 (2020).